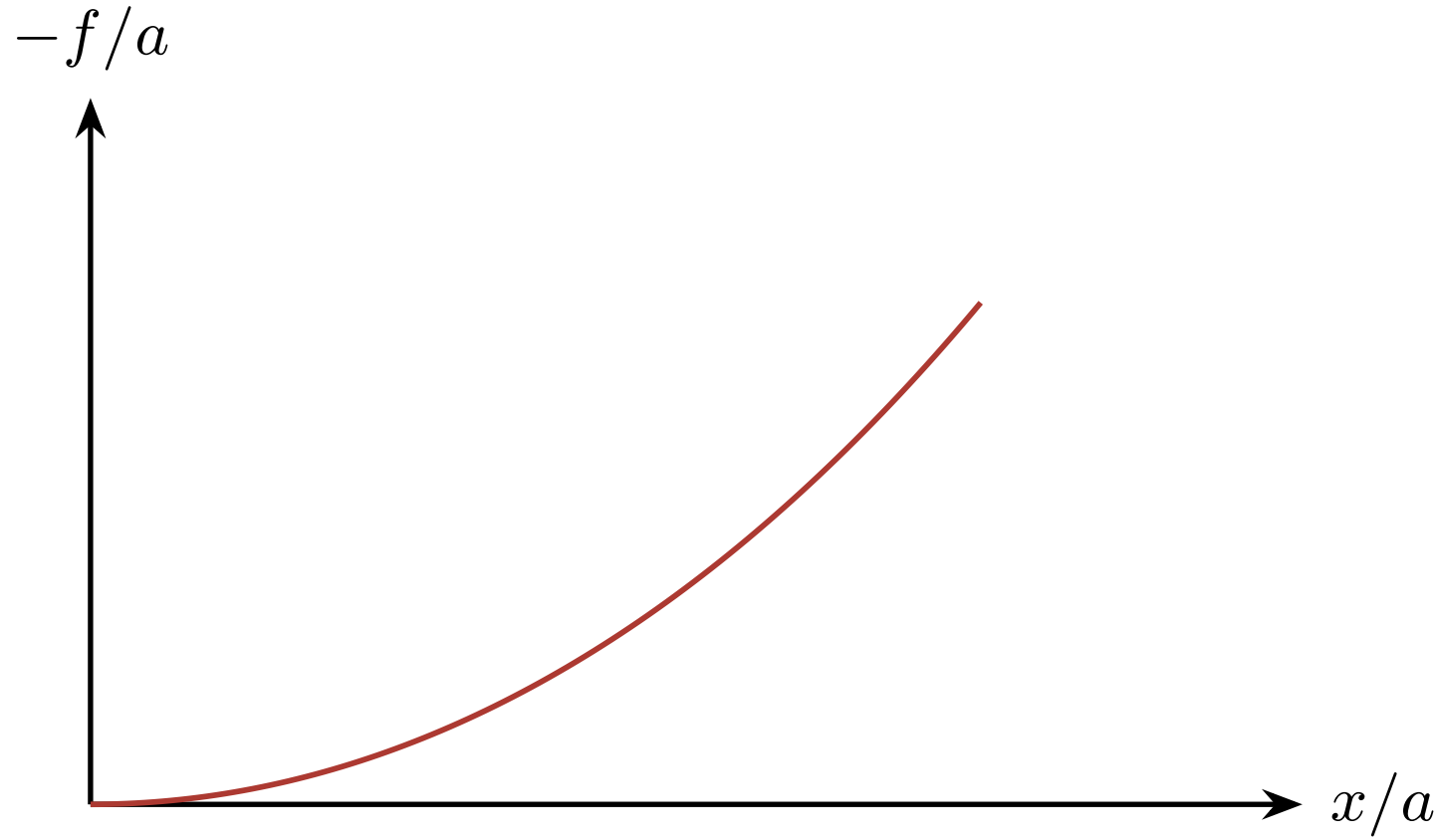

半槽 $f = a[1 - \cosh(x/a)]$；弧长 $s = a\sinh(x/a)$